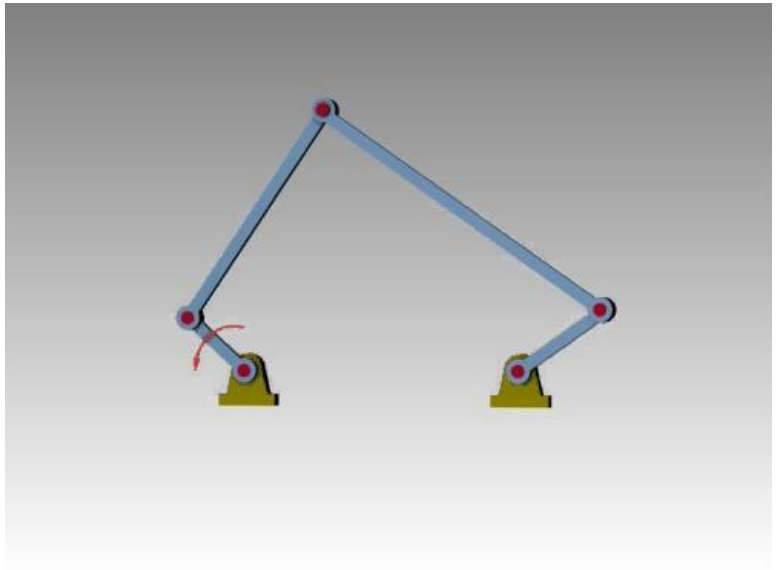
Can it move ?

Can it move with definite motion ?

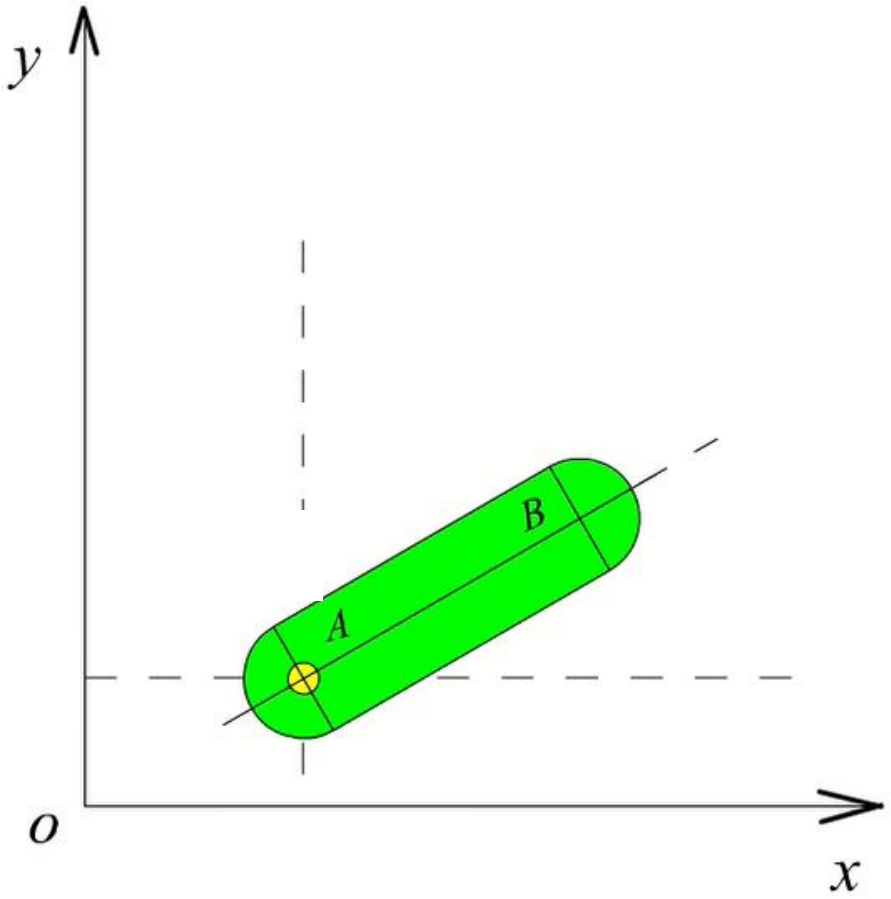


Calculation of the Degree of Freedom of Planar Mechanisms

1.Mater the calculation method

2.Mater how to determine the motion condition based on DOF

1-2 Degree-of-Freedom Calculation for Planar Mechanisms



How many independent motions does a free rigid body have in a plane?

- Translation along x
- Translation along y
- Rotation about z

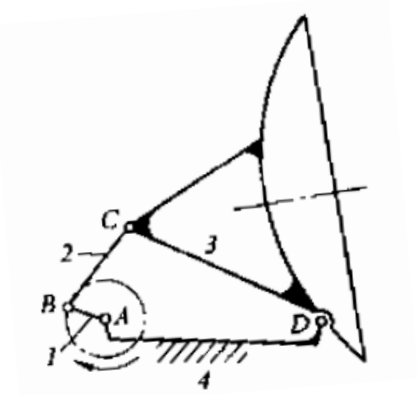
A mechanism consists of several links connected by joints. How to calculate its DOF?

Moving links: n

Lower pair: p_l

Higher pair: p_h

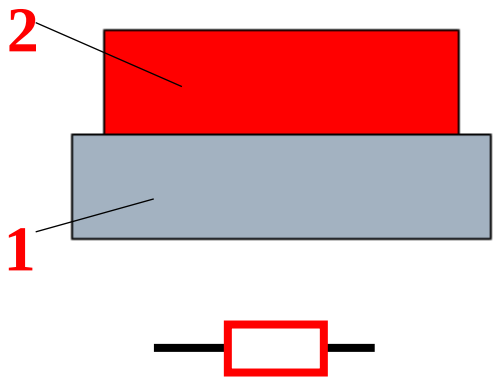
$$F = 3n - 2p_l - p_h$$



Lower pair: surface contact

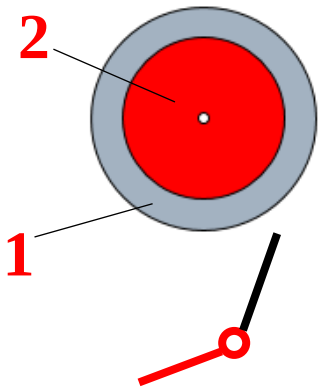
What DOFs it constrains? 2

x, θ_z



P joint

x, y

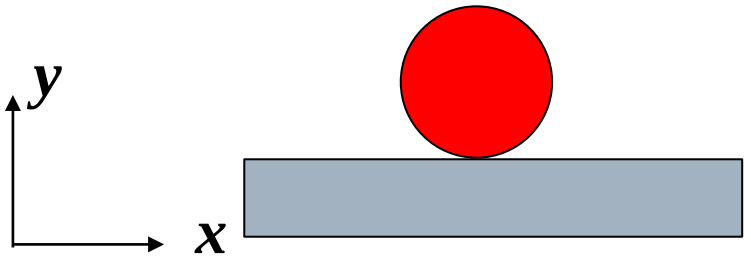


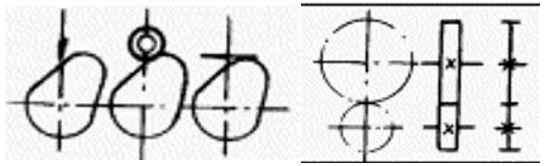
R joint

Higher pair: point or line contact

What DOFs it constrains ? 1

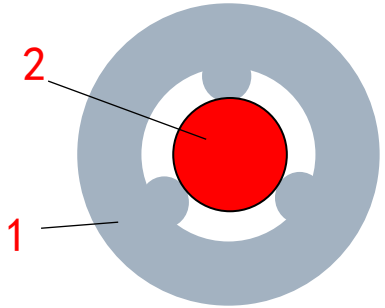
y





Cam Gear

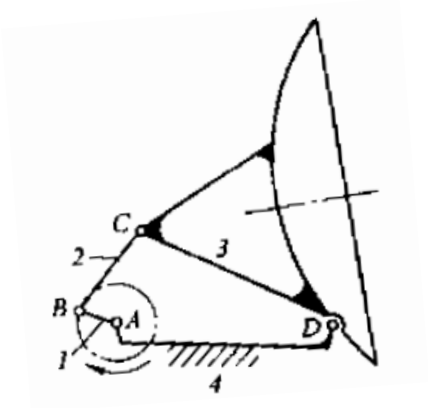
Higher pair?
1 or 3?

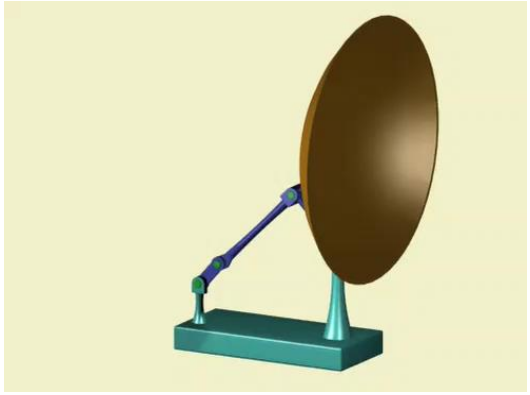


What DOFs it constrains?

2

Calculate its DOF





Moving links: n

Lower pair: p_l

Higher pair: p_h

$$F = 3n - 2p_l - p_h$$

$$F = 3n - 2p_l - p_h$$
$$= 3\times 3 - 2\times 4 - 1\times 0 = 1$$

If this mechanism has one DOF, how many motors do we usually need for definite motion?

If a mechanism has F=1, what does it mean?

That means only one independent input is needed to determine its motion.

How do we judge the motion state of a mechanism?

Compare F with the input number I

Moving links: n

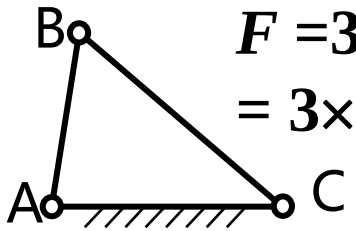
Lower pair: p_l

Higher pair: p_h

$$F=3n-2p_l-p_h$$

1. What if F≤0?

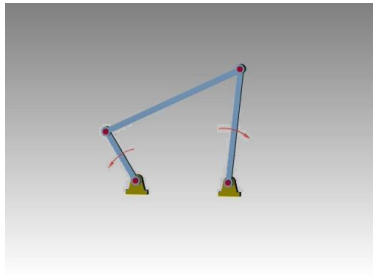
It can't move



$$F=3n-2p_l-p_h$$
$$=3\times2-2\times3-1\times0=0$$

2. What if $F>0$ and $F<I$?

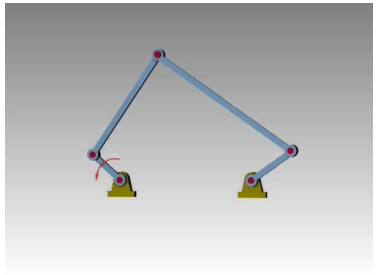
Get stuck or broken



$$F=3n-2p_l-p_h$$
$$=3\times3-2\times4-1\times0=1$$
$$I=2$$

3. What if $F>0$ and $F>I$?

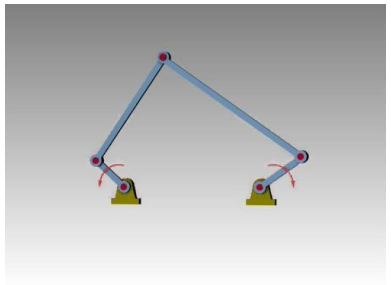
It can move but has indefinite motion



$$F=3n-2p_l-p_h$$
$$=3\times4-2\times5-1\times0=2$$
$$I=1$$

4. What if $F>0$ and $F=I$?

It can move and has definite motion



$$F=3n-2p_l-p_h$$
$$=3\times4-2\times5-1\times0=2$$
$$I=2$$

1-2 Degree-of-Freedom Calculation for Planar Mechanisms

Mechanism Doctor

Diagnose the mechanism:

DOF	Actuators	Motion state
0	1	?
1	1	?
2	1	?
1	2	?

1. Get stuck or broken
2. It can move and has a definite motion
3. It can move but has indefinite motion
4. Get stuck or broken

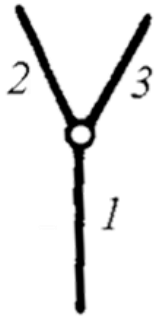
Which case is the healthiest mechanism?

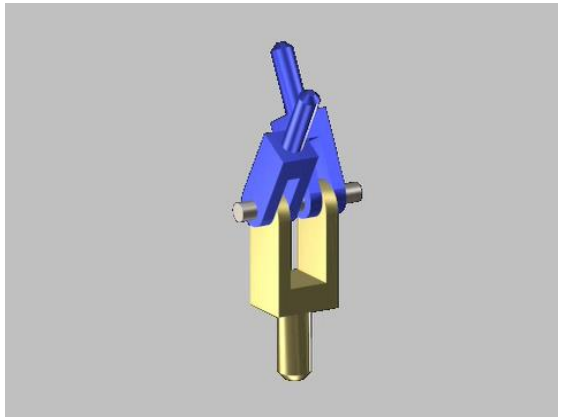
First Trap: Compound Hinge

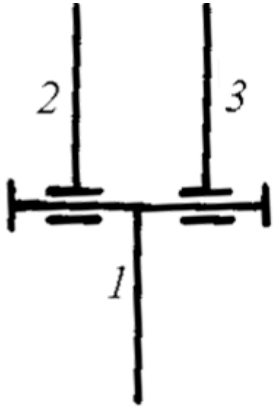
When more than two links are connected at the same point by revolute pairs.

If ***K*** links meet **at one compound hinge**:

Number of revolute pairs=*K*−1







How many R joints?

If 4 links meet at the same hinge, how many revolute pairs should we count?

Second Trap : Local DOF

A local DOF does not affect the overall motion of the mechanism.

Typical example:

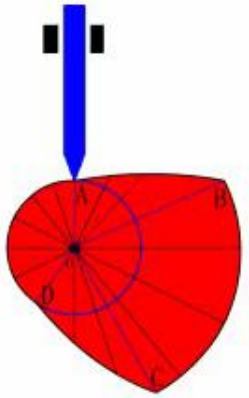
•Roller in a cam mechanism

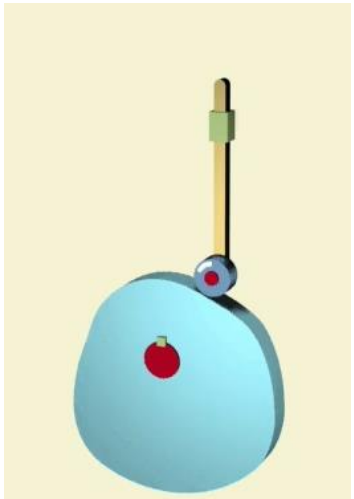
What is roller used for ?

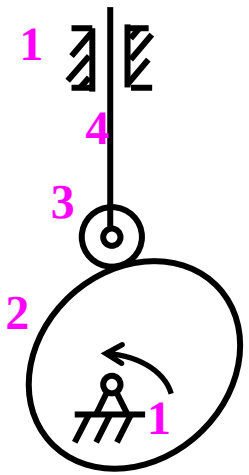
Therefore, be careful when you see a higher pair with a roller

Calculation method:

Treat the roller and its link as one rigid body.





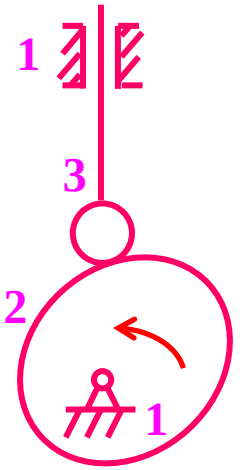


$F = 3n - 2p_l - p_h$

$= 3\times3 - 2\times3 - 1\times1$

$= 2$

Wrong!



$F = 3n - 2p_l - p_h$

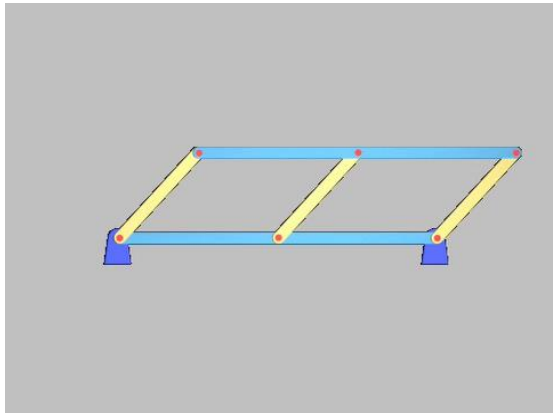
$= 3\times2 - 2\times2 - 1\times1$

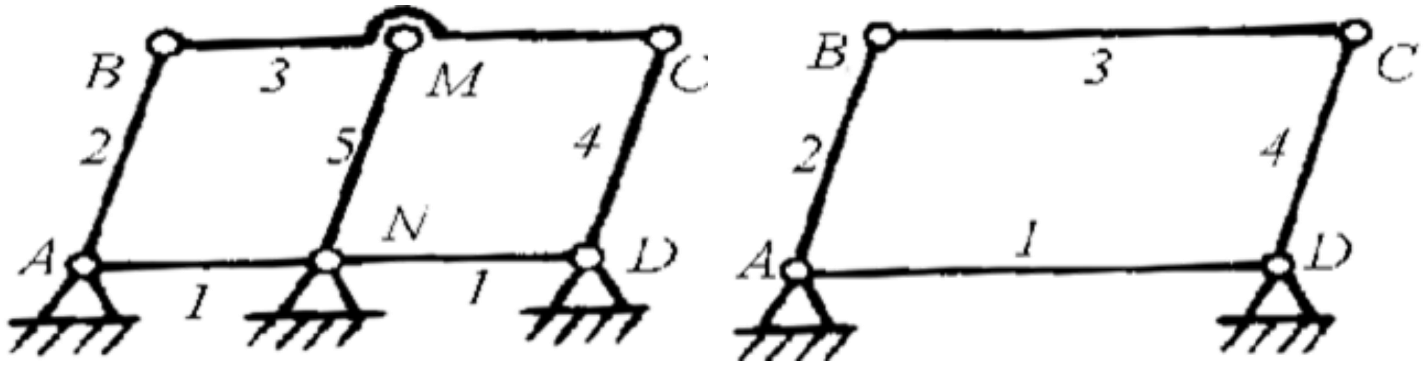
$= 1$

Right !

Third Trap: Passive or Redundant Constraint

Some constraints do not provide independent restriction.
They are repeated constraints.
Before using the DOF formula:
Remove the redundant links and constraints first.





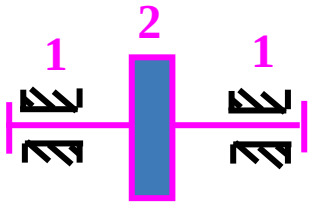
Redundant Constraint Type A: Multiple pairs between two links

Examples:

- **Two revolute pairs with the same axis**
- **Two prismatic pairs with parallel guideways**

Why do engineers sometimes intentionally add redundant constraints?

To improve stiffness, load distribution, accuracy, or reliability.

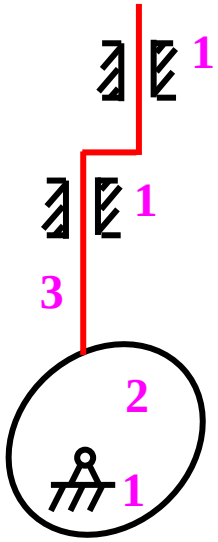


$F = 3n - 2p_l - p_h$

$= 3\times1 - 2\times2 - 1\times0$

$= -1$

Wrong!



Remove the redundant constraints first.

$F = 3n - 2p_l - p_h$

$= 3\times2 - 2\times2 - 1\times1$

$= 1$

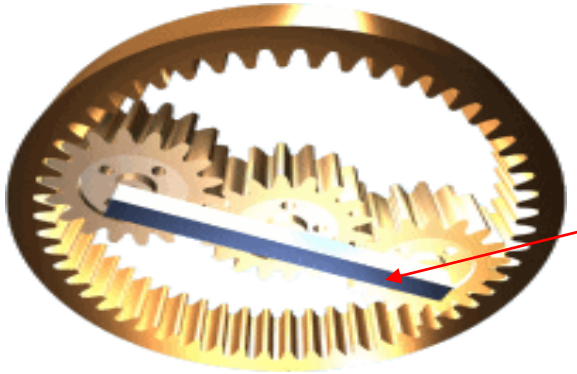
Right!

Redundant Constraint Type B：Symmetric parts not affecting motion

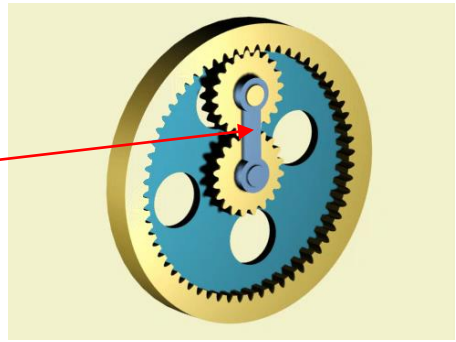
Some symmetric links or contacts do not change the motion pattern.
They may improve:
- **stiffness**
- **stability**
- **load capacity**

But they should not always be counted as independent constraints.

Remove the redundant links and constraints first.



Frame



$F = 3n - 2p_l - p_h$
$= 3\times4 - 2\times4 - 1\times4$
$= 0$

Wrong!

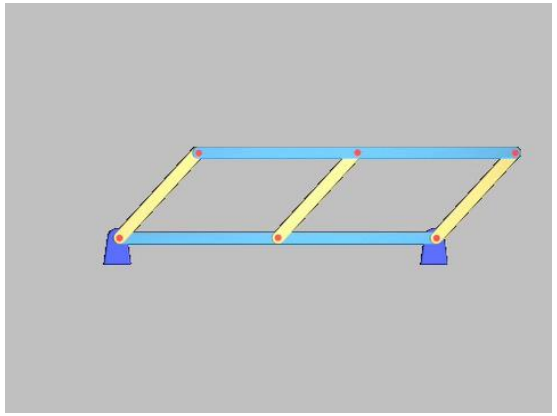
$F = 3n - 2p_l - p_h$
$= 3\times3 - 2\times3 - 1\times2$
$= 1$

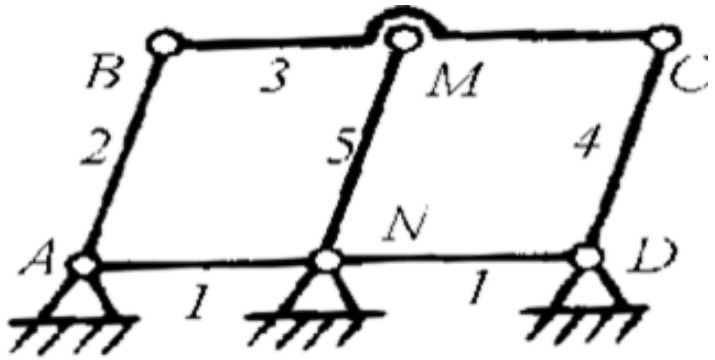
Right!

Redundant Constraint Type C：Constant distance between two points

If the distance between two points remains unchanged during motion, adding a link between them will introduce a redundant constraint.

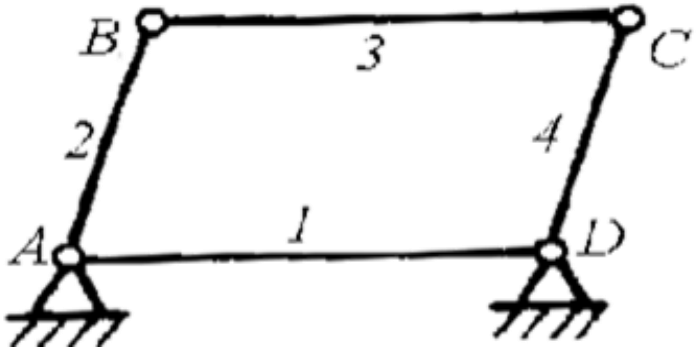
Remove the redundant link and its constraints first.





$$F = 3n - 2p_l - p_h$$
$$= 3\times 4 - 2\times 6 - 0$$
$$= 0$$

Wrong!



$$F = 3n - 2P_L - P_H$$
$$= 3\times 3 - 2\times 4 - 0$$
$$= 1$$

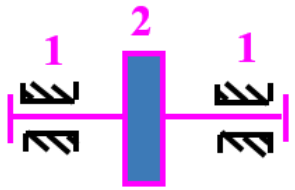
Right!

Why use redundant constraints?



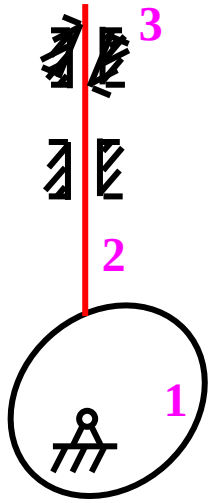
They can:

1.Improve force distribution
2.Increase stiffness
3.Reduce vibration
4.Avoid dead points (talk later)
5.Meet special design requirements



But they require:

•high manufacturing accuracy
•high assembly accuracy
•correct geometric conditions



1.A free planar rigid body has 3 DOFs.

2.Planar mechanism DOF: $F = 3n - 2p_l - p_h$

3.Special cases must be handled carefully:

(a) compound hinge

(b) local DOF

(c) redundant constraints

4.Motion state depends on both DOF and actuator number.

From Formula to Design

DOF calculation helps us answer:

•Can it move?

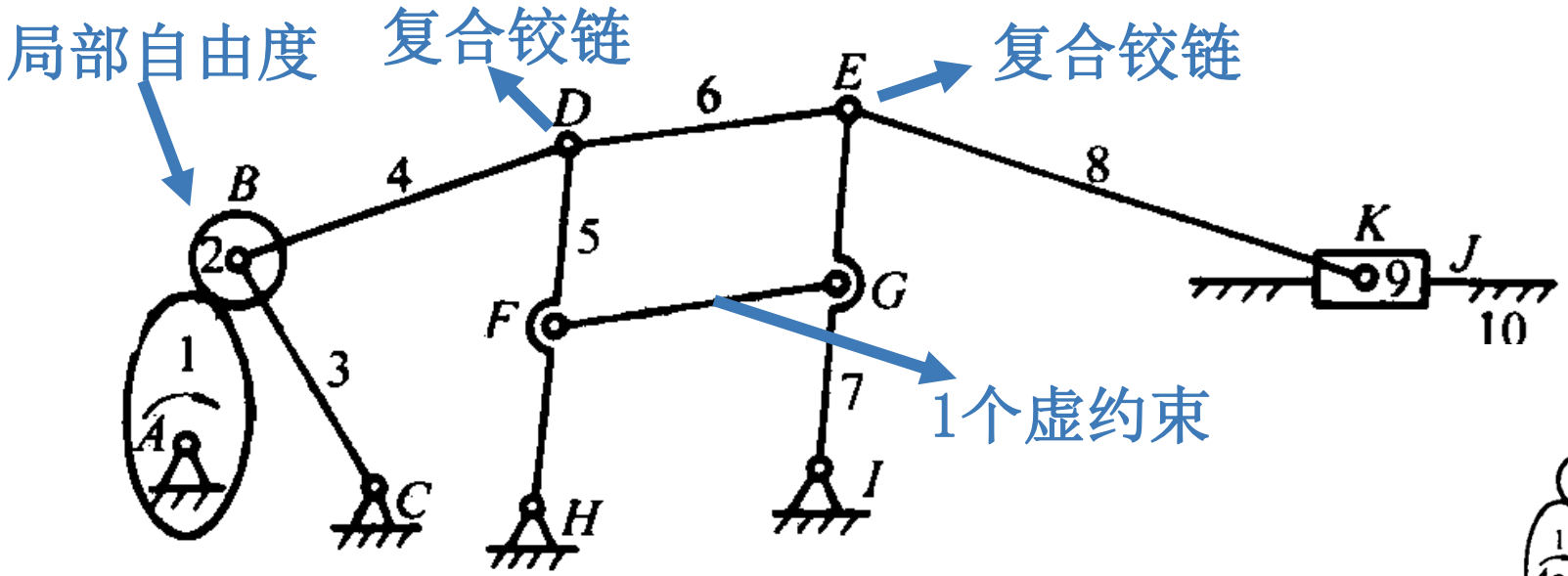
•How many inputs are needed?

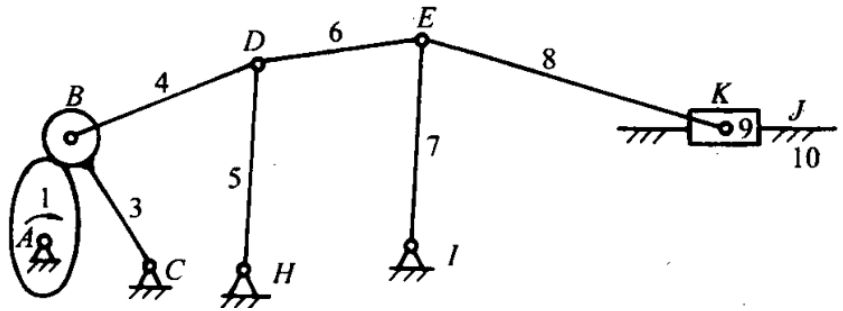
•Is the motion definite?

•Is the design safe and reliable?

1-2 平面机构自由度的计算

练习题：计算图示机构的自由度，如有复合铰链、局部自由度和虚约束，需明确指出。画箭头的构件为原动件。指明机构的运动情况





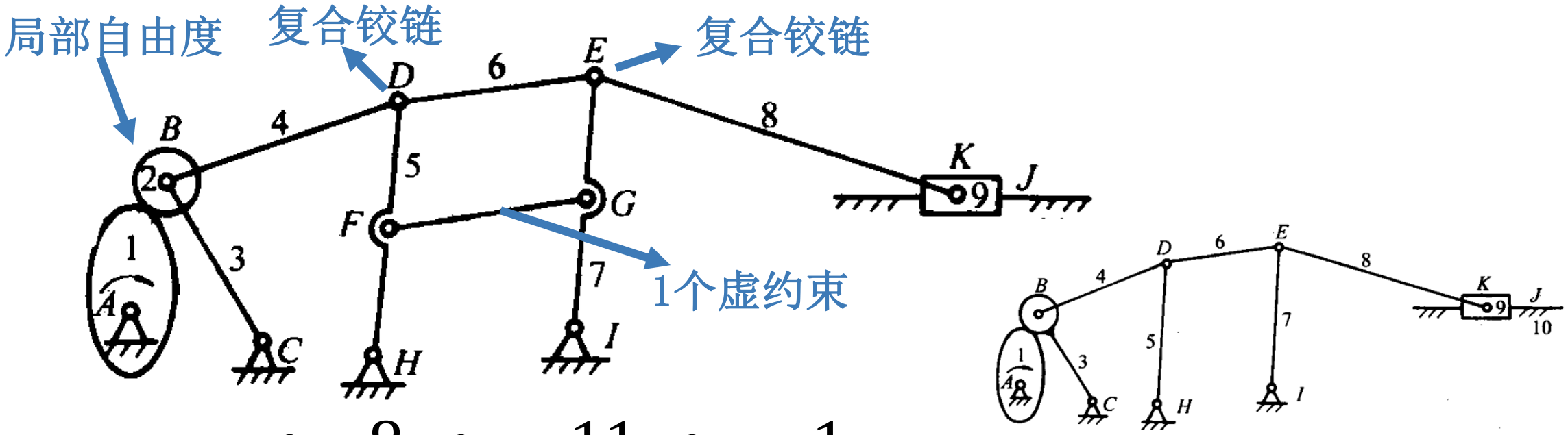
$$n=8,\ p_L=11,\ p_H=1$$

$$F=3n-2p_L-p_H=3\times8-2\times11-1=1$$

F=原动件数，　机构具有确定运动！

1-2 平面机构自由度的计算

练习题：计算图示机构的自由度，如有复合铰链、局部自由度和虚约束，需明确指出。画箭头的构件为原动件。指明机构的运动情况



$$n=8,\ p_L=11,\ p_H=1$$

$$F=3n-2p_L-p_H=3\times8-2\times11-1=1$$

F=原动件数，　机构具有确定运动！